A Literature-Implied Arbitrary-Domain Subcase for Very Weak Monge–Ampère Rigidity in arXiv:2206.09224

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Status and source problem

This is a `literature_implied_answer (partial subcase)` packet. Pakzad proves the full-plane rigidity theorem for $C^{1,\alpha}$ very weak Monge–Ampère solutions when $\alpha > 2/3$, but Remark 1.6 on PDF page 5 records that the analogous result on an arbitrary domain remains open [1].

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Theorem 3. *Assume that $2/3 < \alpha < 1$ and that $f \in \mathcal{D}'(\mathbb{R}^2)$ is a nonnegative nonzero distribution. If $v \in C^{1,\alpha}(\mathbb{R}^2)$ is a solution to $\mathcal{D}et\mathcal{D}^2 v = f$ in $\mathcal{D}'(\mathbb{R}^2)$, then modulo a sign the function v is convex over $\mathbb{R}^2$ and is an Alexandrov solution to $\det \nabla^2 v = \mu_f$ on $\mathbb{R}^2$.*

Remark 1.5. *Note that since $v \in C^1$, we have $|\partial v(A)| = |\nabla v(A)|$ for all Borel sets $A \subset U$.*

Remark 1.6. *A similar result for solutions on a general domain $\Omega \subset \mathbb{R}^2$ was announced by Lewicka and the author in* [25, Theorem 1.4]*. The proof was supposed to appear in a forthcoming paper, but a gap was found. The obstacle is not overcome to this day and the problem for arbitrary domains remains open.*

As it is suggested in Remark 1.4, the main challenging problem is to extend the results of this paper to the case $\alpha > 1/2$. Apart from partial results regarding isometric extensions [8, 4] for this regime, the problem is still largely open and cannot be answered using the current methods.

The extra hypothesis below isolates the missing topological issue: gradient fibers. Under local injectivity, the desired conclusion follows by combining Pakzad's regular-point degree argument with Ball's strict-convexity theorem, in the exact form restated by Guerra and Tione [2, Corollary 5.2].

Partial answer

For $v \in C^1(\Omega)$ and $x \in \Omega$, write

$$v_x(z) = v(z) - v(x) - \langle \nabla v(x), z - x \rangle.$$

Theorem 1 (Locally injective-gradient subcase)**.** *Let $\Omega \subset \mathbb{R}^2$ be a connected open set, let $2/3 < \alpha < 1$, and let $v \in C^{1,\alpha}_{\mathrm{loc}}(\Omega)$ satisfy*

$$\mathrm{Det}\,\mathcal{D}^2 v = f \geq 0 \qquad \text{in } \mathcal{D}'(\Omega).$$

Assume that ∇v is locally injective at every point of Ω. Then there is one sign $\sigma \in \{-1, 1\}$ such that σv is locally strictly convex throughout Ω. If Ω is convex, σv is strictly convex on Ω. Moreover,

$$\mathrm{MA}_{\sigma v} = \mu_f,$$

where μ_f is the Radon measure represented by the nonnegative distribution f and MA *denotes Alexandrov Monge–Ampère measure.*

Why the two cited results fit

Pakzad associates to v its graph $S = \{(x, v(x)) : x \in \Omega\}$ and writes its Gauss map as $N = \eta \circ \nabla v$, where $\eta : \mathbb{R}^2 \to \mathbb{S}^2$ is injective. A graph point is *regular* when its normal is not repeated in a small punctured neighborhood. Thus local injectivity of ∇v makes every graph point regular.

On a relatively compact patch, μ_f is finite. Pakzad's Proposition 3.1 gives bounded extrinsic curvature, and the proof on PDF page 9 identifies the index of a regular point with the local Brouwer degree of ∇v. Corollary 2.2(i) (numbered Corollary 5 in the PDF) makes that degree positive, so every regular point is elliptic [1, pp. 8–9]. In the Pogorelov classification used there, elliptic means that the tangent plane meets a sufficiently small surface neighborhood only at the point itself [4, p. 591]; see also the explicit formulation in [5, p. 430].

$$\deg(w) = \deg(\nabla v, D, \nabla v(x)).$$

Since $\xi_v(B_r(x))$ satisfies (1.1), for all $y \in B_r(x)$, $\nabla v(y) \neq \nabla v(x)$ for $y \neq x$, which finally yields

$$i(p) = \deg(\nabla v, D, \nabla v(x)) = \deg(\nabla v, B_{r/2}(x), \nabla v(x)),$$

since $\nabla v(x) \notin \nabla v(\overline{D} \setminus B_{r/2}(x))$ [19, Corollaire 2.4]. It follows from Corollary 5(i) that any regular point $p \in S$ is elliptic. This implies that S is a surface of nonnegative curvature.

For $z \in \mathbb{S}^2$, and $A \subset S$, let $m_A(z)$ be the cardinality of the set $A \cap \vec{n}^{-1}(z)$. By [33, p. 577, Lemma 3], the function m_A is measurable. It follows then from [33, p. 590, Theorem 3] that the absolute curvature of A equals $\int_{\mathbb{S}^2} m_A(z)\, \mathrm{d}s(z)$, for every Borel subset A of S. On the other hand, for any open set $O \subset S$ and any finite collection of pairwise disjoint closed sets $\{F_i\}_{i=1}^k$ in O, we have for $E_i = \xi_v^{-1}(F_i)$:

$$\sum^k \mathcal{H}^2(\vec{n}(F_i)) = \sum^k \mathcal{H}^2(N(E_i)) \leq \sum^k |\nabla v(E_i)| \leq \mu_f(\xi_v^{-1}(O)).$$

The other input is Ball's theorem. Guerra and Tione state it in exactly the needed low-regularity form: if U is convex, $u \in C^1(U)$, ∇u is locally injective, and $u_{x_0} \geq 0$ near one point x_0, then u is strictly convex on U [2, Corollary 5.2]. Their preceding Theorem 5.1 proves constancy of the critical groups of all affine tilts and gives a new proof of Ball's original result [3].

Corollary 5.2 (Ball [3]). *Assume that Ω is convex and let $u \in C^1(\Omega)$ be such that* D*u is locally injective. If there is $x_0 \in \Omega$ such that*

$$u_{x_0} \geq 0 \text{ in a neighborhood of } x_0, \tag{5.1}$$

then u is strictly convex.

Proof. Since $u_{x_0}(x_0) = 0$, (5.1) asserts that x_0 is a local minimum of u_{x_0}, and in fact it must be isolated: if not, there would be $x_j \to x_0$ with x_j local minima of u_{x_0}, thus we would have $0 = \mathrm{D}u_{x_0}(x_j) = \mathrm{D}u(x_j) - \mathrm{D}u(x_0)$, contradicting the local injectivity of Du at x_0. Hence, by Proposition 4.11(i) we see that $C_k(u_{x_0}, x_0) = \delta_{k,0}\mathbb{Z}$ and so by Theorem 5.1 we must in fact have

Proof of Theorem 1

Fix $x \in \Omega$. Choose a ball $B \Subset \Omega$ containing x on which ∇v is injective. Every point of the graph over B is regular and hence, by Pakzad's result above, elliptic. In particular the tangent plane at $(x, v(x))$ meets a smaller graph neighborhood only at that point. In graph coordinates this says

$$v_x(z) \neq 0 \qquad (0 < |z - x| < r)$$

for some $r > 0$. A punctured planar ball is connected, so v_x has one strict sign there. Consequently x is either a strict local minimum or a strict local maximum of v_x.

If it is a minimum, apply Ball's theorem to v on a smaller convex ball; if it is a maximum, apply it to $-v$. Thus every point has a ball on which one of $v, -v$ is strictly convex. Opposite signs cannot occur on two overlapping balls: on their open intersection the same function would be both strictly convex and strictly concave. Since a connected open subset of $\mathbb{R}^2$ is path connected, a finite chain of overlapping balls along a path shows that one sign σ works throughout Ω.

If Ω is convex, local convexity implies convexity on all of Ω (restrict to line segments). Equality in a convexity inequality along a nontrivial segment would make σv affine on that segment, contradicting strict convexity on a small ball about an interior point. Hence the global convexity is strict.

It remains to identify the measure. Put $w = \sigma v$. On each convex $U \Subset \Omega$, w is convex. Mollify on a slightly larger set. Then w_ε is smooth and convex,

$$\mathrm{MA}_{w_\varepsilon} = \det D^2 w_\varepsilon \, dx = \operatorname{Det} \mathcal{D}^2 w_\varepsilon,$$

and $w_\varepsilon \to w$ locally uniformly. Stability of Alexandrov measures under local uniform convergence of convex functions gives $\mathrm{MA}_{w_\varepsilon} \rightharpoonup \mathrm{MA}_w$. On the other hand,

$$\operatorname{Det} \mathcal{D}^2 w = -\tfrac{1}{2} \operatorname{curl} \operatorname{curl}(\nabla w \otimes \nabla w)$$

and $\nabla w_\varepsilon \otimes \nabla w_\varepsilon$ converges locally uniformly to $\nabla w \otimes \nabla w$, so the same smooth measures converge distributionally to $\operatorname{Det} \mathcal{D}^2 w$. Since $\operatorname{Det} \mathcal{D}^2(\sigma v) = \operatorname{Det} \mathcal{D}^2 v = f$ in dimension two, uniqueness of the limit gives $\mathrm{MA}_w = \mu_f$ on U. Exhausting Ω by such patches completes the proof. $\square$

Provenance, search bounds, and limitation

The packet records a direct implication of known theorems, not a newly claimed Monge–Ampère theorem. Pakzad supplies the nonnegative-degree and elliptic-point statement. Ball supplies the propagation from one local support plane to strict convexity; Guerra and Tione provide a recent, explicit C^1 formulation and proof. Guerra and Tione do not cite arXiv:2206.09224 or state that they answer its arbitrary-domain problem, so the link is an agent-identified literature implication rather than an explicit later answer.

The run's cheap indexes were searched for arXiv:2206.09224, its title, "arbitrary domain," "very weak Monge–Ampère," "locally injective gradient," and the degree/change-of-variables keywords. A bounded web/arXiv search added "gradient local homeomorphism," "gradient index," and "strict convexity Ball," and found arXiv:2506.03906 and Ball's 1980 paper. No exact published statement of Theorem 1 was found.

The original problem remains open because $f \geq 0$ is not known to make ∇v locally discrete or locally injective. The packet does not weaken that added hypothesis. Human review should focus on the localization of Pakzad's elliptic-point conclusion and on the standard Alexandrov-measure limit in the final paragraph.

References

[1] M. R. Pakzad, *Convexity of weakly regular surfaces of distributional nonnegative intrinsic curvature*, J. Funct. Anal. **287** (2024), 110539; arXiv:2206.09224v2, arXiv:2206.09224.

[2] A. Guerra and R. Tione, *Constancy of the index for gradient mappings*, arXiv:2506.03906v2 (2025), Theorem 5.1 and Corollary 5.2, arXiv:2506.03906.

[3] J. M. Ball, *Strict convexity, strong ellipticity, and regularity in the calculus of variations*, Math. Proc. Cambridge Philos. Soc. **87** (1980), 501–513.

[4] A. V. Pogorelov, *Extrinsic Geometry of Convex Surfaces*, Translations of Mathematical Monographs 35, AMS, 1973.

[5] A. A. Borisenko, *Isometric immersions of space forms into Riemannian and pseudo-Riemannian spaces of constant curvature*, Russian Math. Surveys **56** (2001), 425–497.